Paradigmes de Programmation

M1/GL/S1/2025/2026

Série N° 01 (Paradigme fonctionnel)

(Durée: 3H00.)

Exercice 1.

Saisissez chaque ligne suivante dans l'interpréteur OCaml et observez le résultat et le type affiché.

5 ;;	let x = 5 ;;	let l1 = [] ;;	1 - 2 ;;
5. ;;	let y = 10 ;;	let l2 = [1.; 7.; 2.5] ;;	1 - 2 - 3 ;;
5.0;;	x + y;;	let n = (1, 2.) ;;	1 - -2 ;;
'a' ;;	y-x;;	let b = true ;;	1 > 2 - 1 ;;
"bonjour" ;;	x +. y ;;	floor(1.6) ;;	"hello world".[6] ;;
true ;;	let a = "bonjour" ;;	int_of_float(5.1) ;;	Char.code 'a' ;;
(10, 'a') ;;	let b = "les M1" ;;	char_of_int(5) ;;	((((())))) ;;
[1; 2; 7; 5; 7] ;;	a ^ b ;;	float_of_int(5) ;;	(((*1*)))) ;;
1.1e+2 ;;	1 = 2 ;;	int_of_char('5') ;;	(*((()*)) ;;
cos 3.14 ;;		ceil(1.5) ;;	

Exercice 02

Pour chaque expression Ocaml:

- Indiquez si elle est correcte ou incorrecte.
- Si elle est correcte, donnez son type et sa valeur. Sinon, expliquez la cause de l'erreur.

Partie 1	Partie 2
let x = 5 in x ;;	let x = 5 in let x = 7 in x ;;
let x = 5 in x + 2 ;;	let x = 5 in let x = x + 1 in x ;;
let x = 2 + 3 in x * 4 ;;	let x = 5 in let y = x in let x = 10 in x + y ;;
let x = 2 in let y = 3 in x + y ;;	let x = 5 in x + y ;;
let x = 10 in let y = x + 5 in y ;;	let y = 3 in let x = y + 2 in let y = x + 1 in y ;;

Partie 3	Partie 4
let x = 1 and y = 2 in x + y ;;	let x = 2 in let y = x * x in y + 3 ;;
let x = 1 and y = x + 2 in y ;;	let a = 3 in let b = a + 1 in let c = b * 2 in c ;;
let x = 1 and x = 2 in x ;;	let x = 10 in (let y = 5 in x + y) ;;
let a = 1 and b = a + 1 in b ;;	let x = 10 in (let x = 20 in x + 1) + x ;;
let a = 10 and b = 20 in (a, b) ;;	let x = 1 in let y = x + 1 in let x = y + 1 in x + y ;;

Partie 5
let f x = x + 1 in f 5 ;;
let f x = x * x in let y = 3 in f y ;;
let f x = x + 2 in let g x = x * 3 in f (g 2) ;;
let add a b = a + b in add 2 3 ;;
let rec fact n = if n = 0 then 1 else n * fact (n - 1) in fact 4 ;;
let rec somme n = if n = 0 then 0 else n + somme (n - 1) in somme 5 ;;
let rec f x = if x = 0 then 0 else x + f (x - 1) ;;
let f x y = x + y in let g x = x * 2 in f (g 3) (g 4) ;;
let rec f x = f x in f 1 ;;
let f x = let y = x + 1 in y * 2 ;;